

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-3750
Project Name	ComicCraftAI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

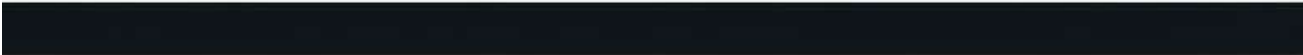
S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited pre-defined art styles	Restricts user creativity and niche appeal	High
2	High latency for high-resolution panels	Negative user experience during creation	Medium
3	Max 5 concurrent users per server	Critical barrier to scaling the service	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Max 5 concurrent users	Transition to a microservices architecture and deploy Auto-Scaling on cloud infrastructure (e.g., AWS EKS).
2	Data Storage	Local server storage for temp files	Integrate a scalable distributed object storage (e.g., AWS S3) with global CDN.
3	Performance	Latency ~5 sec/panel for 4K	Optimize model inference with dedicated GPU instances and implement pre-fetching.
4	Security	Basic user authentication	Implement multi-factor authentication (MFA) and content moderation with adversarial training.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Multi-factor Authentication (MFA)	Q1 2027	Enhance system security and prevent unauthorized access.
Phase 3	Integration with Third-Party Payment Gateways (e.g., Stripe, PayPal)	Q3 2027	Facilitate direct monetization and expand market reach.
Phase 4	Advanced AI-driven Content Moderation Tool	Q1 2028	Automate moderation, reduce operational costs, and improve content quality.